



LINEAR ALGEBRA AND ITS APPLICATIONS

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MATRICES AND GAUSSIAN ELIMINATION

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GAUSSIAN ELIMINATION:

1. Check for consistency and solve the following system of equations if consistent:

$$\begin{aligned} \text{(i)} \quad & x_1 + x_2 - 2x_3 + 3x_4 = 4 \\ & 2x_1 + 3x_2 + 3x_3 - x_4 = 3 \\ & 5x_1 + 7x_2 + 4x_3 + x_4 = 5 \end{aligned} \quad [A:b] = \begin{pmatrix} 1 & 1 & -2 & 3 & : & 4 \\ 2 & 3 & 3 & -1 & : & 3 \\ 5 & 7 & 4 & 1 & : & 5 \end{pmatrix} \begin{matrix} \downarrow \\ R_2 - 2R_1 \\ R_3 - 5R_1 \end{matrix} \rightarrow$$

$$\begin{bmatrix} 1 & 1 & -2 & 3 & : & 4 \\ 0 & 1 & 7 & -7 & : & -5 \\ 0 & 2 & 14 & -14 & : & -15 \end{bmatrix} \begin{matrix} \downarrow \\ R_3 - 2R_2 \end{matrix} \rightarrow \begin{bmatrix} 1 & 1 & -2 & 3 & : & 4 \\ 0 & 1 & 7 & -7 & : & -5 \\ 0 & 0 & 0 & 0 & : & -5 \end{bmatrix}$$

This gives $0 = -5$ which is not possible.

Also $r(A) = 2$ and $r[A:b] = 3$

System is **inconsistent** and has **no solution**

GAUSSIAN ELIMINATION:

$$\begin{aligned} \text{(ii)} \quad & x_1 + 2x_2 + x_3 = 3 \\ & 2x_1 + 5x_2 - x_3 = -4 \\ & 3x_1 - 2x_2 - x_3 = 5 \end{aligned}$$

$$[A:b] = \begin{pmatrix} 1 & 2 & 1 & : & 3 \\ 2 & 5 & -1 & : & -4 \\ 3 & -2 & -1 & : & 5 \end{pmatrix}$$

$$\downarrow \begin{matrix} R_2 - 2R_1 \\ R_3 - 3R_1 \end{matrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & : & 3 \\ 0 & 1 & -3 & : & -10 \\ 0 & -8 & -4 & : & -4 \end{bmatrix}$$

$$\downarrow \begin{matrix} R_3 + 8R_2 \end{matrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & : & 3 \\ 0 & 1 & -3 & : & -10 \\ 0 & 0 & -28 & : & -84 \end{bmatrix}$$

$$\Rightarrow \begin{cases} x_1 + 2x_2 + x_3 = 3 \\ x_2 - 3x_3 = -10 \\ -28x_3 = -84 \end{cases}$$

$r(A)=r[A:b]=3=n$. System is **consistent** and has **a unique solution**.

$$(x_1, x_2, x_3) = (2, -1, 3)$$

GAUSSIAN ELIMINATION:

$$\begin{aligned} \text{(iii)} \quad & 2x - 3y + 2z = 1 \\ & 5x - 8y + 7z = 1 \\ & y - 4z = 3 \end{aligned}$$

$$\begin{pmatrix} 2 & -3 & 2:1 \\ 5 & -8 & 7:1 \\ 0 & 1 & -4:3 \end{pmatrix} \xrightarrow{\substack{R_2 \leftarrow \begin{pmatrix} 5 \\ -8 \\ 7 \end{pmatrix} - \begin{pmatrix} 2 \\ -3 \\ 2 \end{pmatrix} R_1 \\ R_3 \leftarrow \begin{pmatrix} 0 \\ 1 \\ -4 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ -4 \end{pmatrix} R_2}} \begin{pmatrix} 2 & -3 & 2:1 \\ 0 & -1/2 & 2:-3/2 \\ 0 & 1 & -4:3 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 2 & -3 & 2:1 \\ 0 & 1 & -4:3 \\ 0 & -1/2 & 2:-3/2 \end{pmatrix}$$

$$\Rightarrow \begin{cases} 2x - 3y + 2z = 1 \\ -(1/2)y + 2z = -3/2 \end{cases}$$

$r(A)=r(A:b)=2 < n(=3)$ hence system is **consistent**
and has **infinite number of solutions**.
i.e $(x, y, z) = (5k+5, 4k+3, k)$

GAUSSIAN ELIMINATION:

2. Find all values of a for which the resulting linear system has (a) no solution (b) a unique solution and (c) infinitely many solutions:

$$x + y - z = 2$$

$$x + 2y + z = 3$$

$$x + y + (a^2 - 5)z = a$$

$$[A:b] = \begin{pmatrix} 1 & 1 & -1 & : & 2 \\ 1 & 2 & 1 & : & -3 \\ 1 & 1 & a^2 - 5 & : & a \end{pmatrix} \xrightarrow[\begin{smallmatrix} R_2 - R_1 \\ R_3 - R_1 \end{smallmatrix}]{\begin{smallmatrix} R_2 \leftrightarrow R_1 \\ R_3 \leftrightarrow R_1 \end{smallmatrix}} \begin{pmatrix} 1 & 1 & -1 & : & 2 \\ 0 & 1 & 2 & : & -5 \\ 0 & 0 & a^2 - 4 & : & a - 2 \end{pmatrix}$$

- (a) System has **no solution** if $a = -2$ (when $r(A) \neq r(A:b)$)
- (b) System has **a unique solution** if $a \neq \pm 2$ (when $r(A) = r(A:b) = 3 = n$)
- (c) System has **infinitely many solutions** if $a = 2$ (when $r(A) = r(A:b) = 2 < n$)

GAUSSIAN ELIMINATION:

3. Find an equation relating a , b and c so that the linear system
- $$\begin{aligned} x + 2y - 3z &= a \\ 2x + 3y + 3z &= b \\ 5x + 9y - 6z &= c \end{aligned}$$
- is consistent for any values of a , b and c that satisfy that equation. When $(a,b,c)=(2,3,9)$, then what is the solution of the system.

$$\begin{pmatrix} 1 & 2 & -3 & : & a \\ 2 & 3 & 3 & : & b \\ 5 & 9 & -6 & : & c \end{pmatrix} \xrightarrow[\substack{R_2 - 2R_1 \\ R_3 - 5R_1}]{\downarrow} \begin{pmatrix} 1 & 2 & -3 & : & a \\ 0 & -1 & 9 & : & b-2a \\ 0 & -1 & 9 & : & c-5a \end{pmatrix}$$

$$\xrightarrow[\substack{R_3 - R_2}]{\downarrow} \begin{pmatrix} 1 & 2 & -3 & : & a \\ 0 & -1 & 9 & : & b-2a \\ 0 & 0 & 0 & : & c-b-3a \end{pmatrix}$$

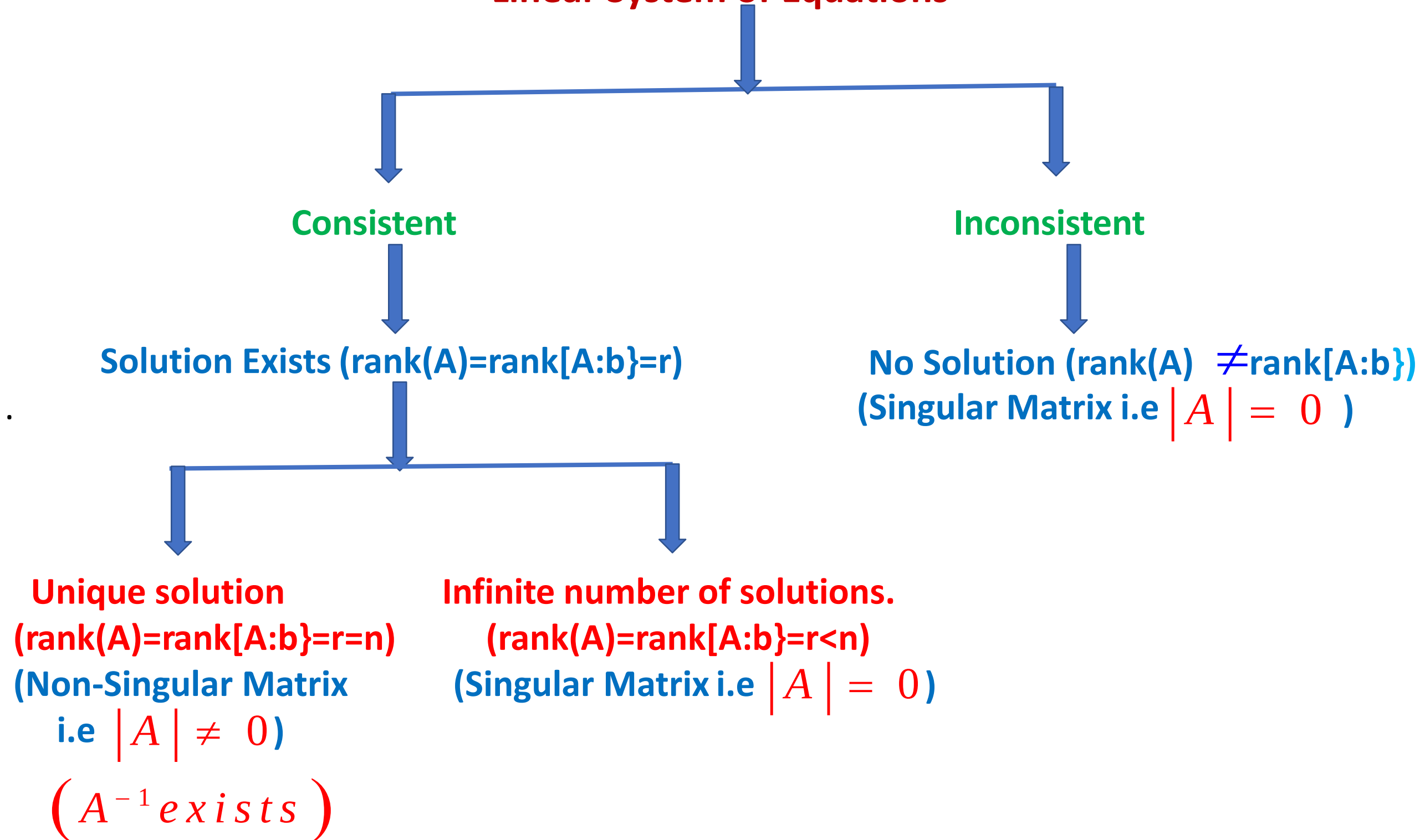
The given linear system will be **consistent** if a, b, c satisfy the relation $c-b-3a=0$.

When $(a, b, c)=(2, 3, 9)$, then the solution of the system is $(x, y, z)=(-15k, 9k+1, k)$

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GAUSSIAN ELIMINATION:

Linear System of Equations





THANK YOU

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